

DAVIDSON COUNTY AP, NC, USA

WMO: 722079

Lat: 35.781N

Lon: 80.304W

Elev: 223

StdP: 98.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 53888

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-5.6	-17.7	0.8	-3.8	-15.0	1.0	-2.3	7.3	12.3	6.4	10.2	0.6	20	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	33.0	23.3	32.4	23.3	31.1	23.0	25.4	29.8	24.9	29.3	24.4	28.6	1.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.8	26.8	23.7	19.1	26.5	23.0	18.3	25.9	79.0	29.9	76.9	29.4	74.8	28.4	33.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
5.8	4.9	4.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-12.0	35.4	3.4	1.5	-14.4	36.5	-16.4	37.4	-18.3	38.2	-20.8	39.3		
			-12.8	26.8	3.0	2.1	-15.0	28.3	-16.7	29.5	-18.4	30.7	-20.6	32.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	4.6	6.3	10.7	15.6	19.8	23.9	25.3	24.7	21.8	15.8	9.5	6.6
	DBStd	8.47	5.67	5.31	5.26	4.34	3.62	2.48	2.09	2.36	3.15	4.43	4.55	4.92
	HDD10.0	574	182	127	59	7	0	0	0	0	0	6	63	131
	HDD18.3	1881	426	337	242	104	29	1	0	0	8	103	266	365
	CDD10.0	2556	16	24	79	173	304	416	474	456	354	186	48	24
	CDD18.3	819	0	1	4	21	74	167	216	198	113	24	1	1
	CDH23.3	6739	0	4	43	221	591	1402	1856	1588	827	192	13	3
	CDH26.7	2323	0	0	4	38	148	499	739	586	272	37	1	0
Wind	WSAvg	1.2	1.3	1.6	1.7	1.7	1.3	1.0	0.9	0.9	0.9	1.0	1.0	1.0
Precipitation	PrecAvg	1107	93	78	108	95	85	104	108	96	99	83	75	84
	PrecMax	1584	178	138	210	210	227	205	190	188	235	338	172	168
	PrecMin	756	27	25	32	12	12	44	49	33	9	0	10	30
	PrecStd	177	39	29	45	48	48	49	42	38	52	74	40	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	23.5	27.2	29.2	32.1	34.0	35.0	36.0	33.1	30.0	25.9	22.7
		MCWB	15.8	17.1	15.8	17.6	21.9	22.7	24.0	22.9	21.6	22.2	17.6	16.6
	2%	DB	18.0	20.4	24.1	27.6	29.9	32.6	33.0	32.8	31.8	27.4	22.6	19.6
		MCWB	14.1	14.7	15.1	17.1	20.9	22.9	24.1	23.6	22.2	19.5	16.3	16.0
	5%	DB	16.3	17.8	22.3	26.1	28.1	31.2	32.2	31.4	29.9	25.8	20.0	17.4
		MCWB	12.6	13.2	14.7	17.1	20.2	22.6	23.9	23.7	22.3	19.1	15.3	14.2
	10%	DB	13.9	15.9	19.9	24.0	27.1	29.6	30.8	29.9	27.9	23.0	17.8	15.1
		MCWB	10.4	11.0	13.8	16.1	19.7	22.5	23.5	23.3	21.0	18.6	13.7	12.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	18.5	18.9	21.5	23.8	25.3	26.2	26.2	25.1	24.0	19.9	19.2
		MCDB	19.2	21.6	22.3	24.5	27.2	30.1	30.4	29.8	28.7	26.6	22.0	20.7
	2%	WB	15.4	16.7	17.5	20.0	22.5	24.6	25.5	25.4	24.1	22.6	18.1	17.2
		MCDB	17.4	18.8	21.8	23.6	26.9	29.4	30.2	29.4	27.7	24.8	20.6	18.6
	5%	WB	13.5	14.6	16.1	18.8	21.7	24.0	25.0	24.9	23.3	21.1	16.4	15.2
		MCDB	15.7	17.2	20.1	23.3	26.2	28.7	29.7	28.9	26.7	23.8	18.8	16.6
	10%	WB	11.2	11.6	14.9	17.7	20.8	23.4	24.4	24.2	22.6	19.6	14.6	12.6
		MCDB	13.5	14.4	19.0	22.2	24.9	27.7	28.9	28.1	25.8	22.7	17.4	14.5
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.3	12.3	12.7	11.5	10.6	10.3	10.0	10.8	12.1	12.7	11.0
		MCDBR	13.6	13.9	14.6	15.0	13.2	12.5	11.9	11.8	13.4	14.0	15.3	13.2
		MCWBR	9.7	9.0	7.5	6.6	5.4	4.5	4.1	4.2	5.0	7.1	9.3	9.4
	5% WB	MCDBR	11.2	12.0	12.1	11.4	10.7	10.3	10.1	9.7	10.2	10.2	11.4	10.3
		MCWBR	9.7	9.2	7.4	5.9	4.9	4.5	4.1	4.2	4.7	6.0	9.0	8.8
Clear-Sky Solar Irradiance	taub	0.311	0.324	0.358	0.398	0.447	0.480	0.529	0.512	0.421	0.379	0.327	0.310	
	taud	2.496	2.455	2.395	2.287	2.183	2.126	2.014	2.044	2.295	2.386	2.491	2.532	
	Ebn at Noon	891	918	910	886	843	812	769	773	834	842	863	871	
	Edh at Noon	82	96	112	131	148	156	173	165	122	102	82	75	
All-Sky Solar Radiation	RadAvg	2.57	3.22	4.29	5.37	5.95	6.38	5.96	5.44	4.57	3.74	2.84	2.20	
	RadStd	0.17	0.39	0.30	0.42	0.49	0.46	0.39	0.26	0.41	0.48	0.23	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+98

Nomenclature: See separate page